Technological Capability, Digital Adoption, and the Internationalization–Performance Relationship: A Firm-Level Study of Singapore

Abstract

This study examines how technological capability (TCI) and digital adoption (DAI) are associated with the internationalization–performance relationship among firms in Singapore, a boundary-case test at the advanced-economy ceiling of the ICRV spectrum, where digital infrastructure is near-universally diffused. Using World Bank Enterprise Survey microdata (Singapore 2023, N = 623), the analysis separates firm-internal capability depth (TCI) from foundational digital interfaces and transaction-enabling mechanisms (DAI).

Three findings emerge. First, the I–P relationship within the observed export-intensity range is better characterized as predominantly positive with mild quadratic curvature than as an inverted-U; the implied turning point lies in a sparsely populated upper tail. Second, TCI is positively associated with labour productivity, but no TCI moderation of the I–P curve is detected. Third, DAI does not produce a uniform productivity premium; instead, its productivity association becomes more positive only at higher export intensity, with the clearest signal in the high-export tail. These findings suggest that foundational digital adoption functions as a contingent scaling resource, its productivity relevance activates under cross-border coordination demands, rather than as a firm-wide productivity advantage.

Keywords: internationalization–performance relationship; digital adoption; technological capability; export intensity; labour productivity; Singapore

1 Introduction

1.1 Background and motivation

Singapore is an analytically valuable setting for revisiting the internationalization–performance relationship because it combines high digital maturity with firm-level heterogeneity in export intensity. The classic I–P literature explains nonlinearity through a trade-off between the scale and learning benefits of international expansion and the rising coordination costs of operating across markets, with the inverted-U emerging as the modal empirical pattern in the broader literature (Marano et al. 2016). However, much of this literature was developed in pre-digital or transitional-digital settings, where institutional and infrastructural conditions differ substantially from those of digitally advanced economies (Peng 2003; Peng et al. 2008) and may not fully capture how the I–P curve behaves when firms operate in a digitally advanced institutional environment. These conditions are empirically grounded in contemporaneous assessments: Singapore ranked 2nd globally in the StartupBlink Innovators Business Environment Index 2026 (score 99.145 out of 100; 1st in Asia Pacific), 4th in the Global Startup Ecosystem Index 2025 (up from 5th in 2024, with ecosystem growth of 44.9% against a top-20 average of 28.5%), and 3rd in the United Nations e-Government Survey 2024, while the Smart Nation 2.0 initiative (launched 2024) formalizes digital trust infrastructure and technology-enabled growth as explicit national policy priorities (GovTech Singapore, 2024; StartupBlink, 2025, 2026). In such contexts, the relevant question is not simply whether firms still exhibit the conventional inverted-U, but whether digital maturity changes the export range over which the right-side decline becomes empirically visible. If mature digital infrastructure attenuates communication, transaction, and information frictions, then the coordination-cost mechanism underlying the classic inverted-U may be weakened, delayed, or pushed toward sparsely populated levels of export intensity. This possibility is especially important in environments where firms engage with digital interfaces and transaction systems at scale, because such technologies can alter both the cost and speed of cross-border coordination (Bharadwaj et al. 2013; Verhoef et al. 2021). This combination creates what we term a *digital saturation paradox*: in an environment where Tier 1–2 digital interfaces (websites, electronic payment systems) are already widely diffused, these tools appear to function as hygiene factors for domestic operations rather than differentiating assets. This near-universal Tier 1–2 diffusion is enabled by Singapore's nationally deployed Digital Public Infrastructure, encompassing digital identity systems, payment rails, and data-exchange platforms, which the World Bank (2025a) identifies as the foundational enabling layer for firm-level digital adoption in advanced economies. Consistent with this, the Singapore sample reveals that 82% of firms report zero exports, while only 3% exceed 50% foreign sales to total sales, a striking domestic orientation despite the economy's mature digital infrastructure. As a result, the productivity returns to foundational digital adoption should not be expected to manifest uniformly across sampled firms; instead, they should become measurable primarily when firms face the intensified coordination and transaction demands of cross-border activity. Singapore therefore serves not merely as a convenient empirical backdrop, but as an analytically ideal stress test an environment in which the conditional logic of foundational digital adoption is most legible precisely because domestic saturation removes the uniform-premium effect and leaves only the export-contingent signal. A second motivation concerns construct clarity. In digitally advanced settings, firms may differ not only in deeper technological capability rooted in learning, innovation, and absorptive capacity, but also in the extent to which they adopt basic digital interfaces and transaction-enabling systems. These two domains should not be collapsed into a single umbrella construct, because technological capability reflects firm-internal capability depth grounded in resource-based and absorptive-capacity logic (Barney 1991; Cohen & Levinthal 1990; Lall 1992), whereas digital adoption reflects participation in digitally enabled transactions and interfaces that may be more contingent on the surrounding digital ecosystem (Bharadwaj et al. 2013; Verhoef et al. 2021).

1.2 Research gap

Three gaps motivate this study. First, prior research has often bundled digitalization-related firm attributes into broad umbrella constructs, making it difficult to distinguish firm-internal technological capability from more basic forms of digital adoption. This matters because the two domains imply different mechanisms of advantage: technological capability concerns learning, innovation, and absorptive depth within the firm, whereas foundational digital adoption concerns the use of digital interfaces and transaction-enabling systems that may matter differently across stages of internationalization. Second, although work on digital internationalization has generated important conceptual advances, firm-level evidence remains limited on how basic digital adoption relates to export intensity within a single digitally advanced institutional environment. In particular, the literature still offers limited evidence on whether foundational digital tools matter uniformly across firms or become more relevant only when cross-border coordination demands intensify. Focusing on Tier 1–2 adoption, the most widely diffused and comparable digital layer, is therefore a deliberate design choice: if even these foundational tools exhibit export-contingent productivity associations, higher-tier technologies (automation, integrated ERP, AI-enabled systems) should produce even stronger differentiation, making this the most conservative and replicable test of the conditional digital adoption hypothesis. Third, the nonlinear internationalization–performance literature has documented inverted-U patterns extensively, but the interpretation of such patterns in a digitally advanced setting remains underspecified. In a case such as Singapore, the unresolved issue is not whether one setting can establish a general boundary condition for all digitally mature economies, but how firm-level evidence from that setting should be interpreted relative to the conventional nonlinear literature. This study addresses that issue by examining whether the right-side decline of the curve is clearly identifiable within the export-intensity range actually occupied by firms in Singapore and by separating technological capability from foundational digital adoption in the same empirical framework.

1.3 Contribution

This study makes one primary contribution and two supporting contributions. Its primary contribution is to document that foundational digital adoption (DAI) functions as a conditional scaling resource rather than a uniform productivity premium in a digitally advanced economy: the productivity association of Tier 1–2 digital adoption is weak across most of the export-intensity distribution and becomes more positive only in the high-export tail, where cross-border coordination demands are densest. This within-context evidence from Singapore supports a contingent digital complementarity mechanism, DAI as a scaling lever whose value is realised under export-intensity conditions rather than as a broad firm-level capability advantage. The evidence for this mechanism (H3: positive quadratic DAI×$FSTS^2$ interaction, β=+3.119, p=.005) is the most distinctive empirical finding and the one least anticipated by prior literature. The first supporting contribution qualifies the conventional nonlinear I–P literature. Within the observed export-intensity range, the fitted quadratic is more defensibly read as predominantly positive with mild curvature than as a formally established inverted-U: the implied turning point lies near FSTS=88.6%, in a sparsely populated upper tail, and the Lind–Mehlum test does not formally confirm a right-side decline at conventional thresholds. In a digitally mature economy where Tier 1–2 coordination tools are widely diffused, this is an informative positive null, consistent with the saturation hypothesis, rather than an anomaly. The study therefore qualifies rather than overturns the nonlinear I–P literature. The second supporting contribution is construct clarification. The analysis distinguishes technological capability (TCI) from foundational digital adoption (DAI), separating firm-internal capability depth from digitally enabled interfaces and transaction mechanisms. This distinction matters because the two constructs are associated with firm performance through different empirical patterns: TCI shows a more stable direct productivity association, whereas DAI shows a contingent export-intensity-dependent pattern.

The Section 4.5 R1 indicator-sensitivity diagnostic further clarifies that the Tier-1+2 composition of DAI combining website presence with two-way electronic-payment intensity is what generates the discriminatory power across the export-intensity range; a Tier-1-only restriction substantially weakens the quadratic interaction, indicating that the empirical legibility of the conditional-scaling mechanism depends on the institutional availability of Tier-2 transaction-enabling infrastructure (see Section 5.1 for the construct-boundary discussion).

1.4 Roadmap

Section 2 develops the theoretical framework and four hypotheses. Section 3 describes data and methods. Section 4 reports results. Section 5 discusses theoretical and managerial implications. Section 6 concludes. Section 7 acknowledges limitations and future research directions.

2 Theory and Hypotheses

2.1 The internationalization–performance relationship

The I–P literature has produced extensive evidence of nonlinear relationships between the degree of internationalization and firm performance. Hitt et al. (1997) introduced the inverted-U logic: initial internationalization yields scale and learning benefits, but at high export intensities firms encounter cognitive and coordination ceilings that erode the returns to further expansion. Contractor et al. (2003) extended this logic to a three-stage S-curve, and Lu and Beamish (2004) refined the inverted-U with attention to firm size and age effects. The modal pattern across this literature is an inverted-U with a turning point in the 30%–60% foreign-sales-to-total-sales (FSTS) range (Marano et al. 2016). An important implication of the nonlinear I–P literature is that the visibility of an inverted-U depends not only on theory but also on where firms actually lie in the observed distribution of internationalization intensity. If firms operate in a setting where digital infrastructure and transaction systems reduce some coordination frictions, the right-side decline may become harder to detect within the export range most firms occupy, even if coordination costs do not disappear in principle. Singapore is therefore analytically useful not because it can by itself establish a general boundary condition for digitally mature economies, but because it provides a within-context case for examining how the conventional I–P logic appears when firms operate in a digitally advanced institutional environment.

2.2 Two distinct constructs: TCI and DAI

A recurring problem in the digital international business literature is the tendency to collapse technologically heterogeneous firm attributes into a single umbrella construct, thereby blurring the distinction between internal capability depth and digitally enabled market participation (Verhoef et al. 2021). This study departs from that practice by distinguishing a Technological Capability Index (TCI), anchored in the Lall–Cohen-Levinthal capability tradition (Cohen & Levinthal 1990; Lall 1992), from a Digital Adoption Index (DAI), anchored in the Bharadwaj–Verhoef digitalization tradition (Bharadwaj et al. 2013; Verhoef et al. 2021). The distinction is theoretically necessary because the two constructs refer to different domains of firm advantage. TCI captures internal capability stocks related to innovation, learning, and technology absorption, drawing on the absorptive-capacity logic of Cohen and Levinthal (1990) and the technological-capability tradition of Lall (1992). DAI, by contrast, captures the firm's uptake of digital interfaces and transaction-enabling mechanisms (Bharadwaj et al. 2013), reflecting a more environmentally contingent form of firm advantage that depends on the maturity of the surrounding digital ecosystem (Verhoef et al. 2021). Equally important, the label "digital adoption" is more defensible than "digital capability" in the present empirical setting because the WBES indicators available here map primarily onto basic digital presence and digital transaction usage rather than deeper digitally integrated organizational processes. In the Verhoef et al. (2021) hierarchy, website presence (c22b) corresponds most closely to Tier 1 digitization, while electronic-payment intensity on the customer and supplier sides (k33, k38) corresponds to Tier 2 digitalization. Our DAI composite therefore captures the foundational digital layer upon which higher-order digital capabilities may be built, but it cannot directly observe Tier 3 process integration (e.g., ERP, CRM, digitally integrated supply chains) or Tier 4 digital dynamic capability (e.g., AI deployment, platform orchestration, reconfiguration capability). Given this measurement boundary, retaining a broad "digital capability" label would overstate what the underlying indicators can legitimately support, whereas the DAI label improves construct validity by aligning the variable name with the actual content domain of the observed measures. Methodologically, the two constructs satisfy the Coltman et al. (2008) four-criterion test for formative (as opposed to reflective) measurement. Separately, combining TCI and DAI into a single composite would violate the principle that constructs with different hypothesised nomological nets should remain analytically distinct.

2.3 Hypothesis development

2.3.1 TCI direct effect on productivity Following the Lall (1992) capability tradition and the Cohen and Levinthal (1990) absorptive-capacity logic, firms with deeper technological capability should produce more output per unit of labour input regardless of internationalization stage. Technological capability reflects firm-internal investments in R&D, innovation, foreign technology licensing, and quality certification, all of which are associated with a higher productivity floor, reflected in operational efficiency, product quality, and learning routines (Avenyo et al. 2021; Krammer et al. 2018). We therefore hypothesize: Hypothesis 1 (H1). In Singapore's advanced-economy context, when firms invest more deeply in R&D, innovation, technology licensing, and quality certification, labour productivity is higher because technological capability raises the firm's operational efficiency and learning routines (Lall 1992; Cohen & Levinthal 1990), before any additional moderation of the I–P curve shape is considered.

2.3.2 TCI moderation of the I–P relationship A subtler prediction concerns whether technological capability alters the shape of the internationalization–performance relationship rather than simply raising average productivity. From a non-location-bound firm-specific advantage perspective, technological capability may travel across markets without requiring the same degree of adaptation as context-dependent coordination tools. This suggests that its most stable empirical role may lie in its direct association with productivity, while any moderation of the FSTS–performance relationship is less certain and should be treated as an empirical question rather than presumed ex ante. Whether technological capability (TCI) also moderates the internationalization–performance relationship is treated as an open empirical question and assessed in a supplementary specification rather than as a hypothesized moderation channel. Consequently, this paper formally states H1, H2, and H3 only; a TCI moderation hypothesis is not posited for the Singapore context, where the advanced-economy setting and near-universal digital diffusion do not generate the institutional heterogeneity that would support a confident ex-ante prediction about TCI-contingent I–P curvature. The present paper's H2 and H3 are accordingly numbered consecutively after H1: H2 captures the conditional (non-uniform) DAI–productivity association, and H3 captures the export-intensity-contingent moderation of DAI on the I–P relationship.

2.3.3 DAI direct effect on productivity Digital adoption is associated with firm performance through faster information flows, lower transaction frictions, and more efficient digital exchange, but such associations are unlikely to be uniform across firms. In the present empirical setting, the available indicators capture foundational digital interfaces and transaction-enabling mechanisms rather than deeper digitally integrated organizational capabilities. Their productivity relevance is therefore not expected to appear as a large, unconditional firm-wide premium, but as an association whose strength may depend on the scale at which firms actually use those digital channels. Hypothesis 2 (H2). In Singapore, when firms adopt foundational digital interfaces and transaction-enabling mechanisms (Tier 1–2), the productivity association of DAI is conditional rather than uniform across firms, because the marginal return to basic digital adoption depends on the scale at which those digital channels are actually used (Bharadwaj et al. 2013; Verhoef et al. 2021), until a level of export intensity is reached at which cross-border coordination demands generate measurable returns.

2.3.4 DAI as a conditional complement to export intensity The most specific prediction concerns whether foundational digital adoption becomes more relevant when firms face denser cross-border coordination demands. Firms with higher export intensity process more transactions across customers, suppliers, and markets, making digital interfaces and electronic transaction systems potentially more valuable as scaling mechanisms. By contrast, firms with low export intensity may have fewer opportunities to translate basic digital adoption into measurable labour-productivity gains.

Three candidate mechanisms underpinning the export-contingent DAI–productivity association. Before stating H3, we name the three theoretical channels that could generate a positive export-contingent DAI association, because identifying the focal mechanism in advance disciplines the empirical reading of the interaction terms and the robustness diagnostics. *Mechanism 1, Substitution*: foundational digital adoption compensates for weaker alternative coordination resources at higher export intensity by reducing per-unit cross-border coordination cost, thereby raising the productivity of high-intensity exporters relative to low-DAI peers. Substitution predicts a *level shift* of the I–P curve for high-DAI firms but does not necessarily require accelerating returns to DAI as export intensity rises. *Mechanism 2, Amplification*: foundational digital adoption raises the *marginal* productivity return to each additional unit of export intensity, effectively rotating the I–P curve upward for high-DAI firms rather than shifting it in parallel. Amplification predicts an accelerating contribution of DAI as export intensity rises and is empirically distinguishable from substitution through the sign and significance of the *quadratic* FSTS × DAI interaction: a positive quadratic coefficient is consistent with amplification because it implies that DAI's productivity contribution grows super-linearly with export-intensity scale, which a level-shift substitution mechanism would not produce. *Mechanism 3, Selection*: inherently more productive firms simultaneously select into high digital adoption and high export intensity, producing a positive cross-sectional interaction in the absence of any causal scaling mechanism. Selection predicts attenuation under sample restrictions or sensitivity diagnostics that remove the firms driving the simultaneous selection, and is the principal identification threat in a single-wave cross-section.

H3 is therefore framed under the *amplification* expectation: the productivity-relevant role of foundational digital adoption is theorised to *accelerate* with export intensity through the quadratic FSTS × DAI interaction, rather than to produce a level shift (substitution) or a selection-driven artefact. This commitment to amplification as the focal mechanism is important for interpretation, because the positive *quadratic* (not just linear) interaction term will serve as the empirical test of the amplification prediction in Section 4. The substitution and selection mechanisms are retained as alternative explanations to be assessed against the robustness evidence in Section 4.5.

Hypothesis 3 (H3). In Singapore, when firms operate at higher export intensity, the association between digital adoption (DAI) and firm performance becomes more positive because foundational digital tools reduce cross-border coordination costs super-linearly rather than proportionally (coordination platform mechanism: Stallkamp & Schotter 2021; Barney 1991), with the amplification effect becoming detectable only beyond the export-intensity threshold at which transaction volume justifies intensive use of digital interfaces and electronic-payment systems.

2.4 Conceptual model

Figure 1 summarizes the conceptual model. The model treats TCI and DAI as analytically distinct firm-level constructs that are associated with firm performance through different channels. TCI enters the model primarily as a direct-effect construct, reflecting firm-internal capability depth grounded in learning, innovation, and technology absorption; whether TCI also moderates the FSTS–performance relationship is examined in a supplementary test rather than presumed ex ante. DAI, by contrast, is the contingency variable on the I–P path: its productivity relevance is theorized to vary across levels of export intensity rather than to operate as a uniform direct premium, because the observed indicators capture foundational digital interfaces and transaction-enabling mechanisms whose relevance depends on how intensively firms engage in cross-border operations. In this sense, H2 and H3 are jointly informative: H2 concerns the non-uniformity of the DAI–productivity association, whereas H3 specifies the direction of that contingency across export intensity. The model is therefore positioned as a within-context framework for interpreting firm-level evidence from Singapore rather than as a design that can, on its own, establish a general boundary condition for digitally mature economies.

Figure 1: Conceptual model, TCI and DAI moderating the I–P relationship (Singapore)

Figure 1. Conceptual model for Paper 4 (Singapore).

Note: Solid arrows represent hypothesised direct effects; dashed arrows represent moderating and conditional effects. The independent variable, internationalisation intensity (FSTS, FSTS²_c), is placed at left; the dependent variable, ln(labour productivity), is placed at right. H1 (inverted-U tendency with support-constrained turning point ~88.6% FSTS, bootstrap CI [53%, 253%]) is grounded in coordination-cost theory (Hitt et al., 1997); the right-side decline is not formally identified within the observed FSTS range, a theoretically informative result under the saturation framework (Lind–Mehlum $p = .303$). H1-TCI: technological capability (TCI_z, RBV: Barney, 1991) enters as a direct level-shift to firm performance. H2: DAI (Tier-1+2: website + e-payment) exhibits a non-uniform positive direct effect, varying across export intensity. H3: DAI amplifies I to P performance returns at high FSTS, the primary contingency hypothesis (coordination platform mechanism: Stallkamp & Schotter, 2021; Dynamic Capabilities: Teece, 2007). The Heckman two-step IMR sensitivity check controls exporter selection bias but is not displayed in the figure. (+) denotes positive association. Control variables (ln firm size, age, foreign ownership, sector FE) are included in all empirical models. Note: H3 amplification test conducted on exporters-only subsample ($n \approx 84$; power $\approx 16\%$, corroborating signal, not independent identification). ICRV context: Singapore, Advanced I. Data: WBES 2023; $N = 623$.

3 Methods

3.1 Data

We use the World Bank Enterprise Survey (WBES) Singapore 2023 wave, the most recent firm-level microdata available for Singapore. The 2023 wave was administered under the B-READY methodology, which expanded the digital-adoption module to include items on electronic-payment penetration. After listwise deletion on focal variables, the analytic sample comprises 623 firms across major sectors of the Singapore economy. Manufacturing accounts for approximately 31% of the sample, retail and other services for 50%, and other sectors for

19%. The sample is predominantly domestic in orientation: 18% of firms report non-zero exports, and only 3% exceed 50% foreign sales to total sales (FSTS). This domestic skew is not anomalous for the WBES sampling frame, which targets privately held firms with fewer than 300 employees across manufacturing and services sectors, and therefore systematically under-represents the large, export-oriented multinationals that account for the majority of Singapore's aggregate trade flows (national trade/GDP exceeds 300%). The sample is accordingly representative of Singapore's SME sector rather than of the national export economy; external-validity inferences should be bounded to firms of comparable size and domestic orientation. Descriptive statistics and pairwise correlations are reported in Table 1.

DAI distribution note. The Singapore DAI distribution reveals high saturation: website presence is reported by approximately 67% of firms, and Tier-2 e-payment adoption (customer- and supplier-side) is widespread relative to less digitally mature institutional settings. This saturation pattern at the advanced-economy ceiling of the digital-infrastructure spectrum has two empirically relevant implications. First, the near-universal diffusion of Tier-1 digital presence removes the discriminatory power that the website indicator alone could carry in less digitally mature settings, so the empirical signal must be drawn from the Tier-2 transaction-enabling layer (e-payment intensity on both customer and supplier sides) to identify the conditional-scaling mechanism, a measurement implication that is borne out in the Section 4.5 R1 robustness diagnostic, where reducing DAI to Tier-1 alone substantially weakens the quadratic interaction. Second, in transitional-economy settings where Tier-1 digital adoption has diffused widely while Tier-2 transaction-enabling infrastructure remains institutionally bounded, the same construct measured at Tier-1 only would not be expected to produce a comparable positive quadratic interaction; instead it would be more plausibly read as construct-tier obsolescence than as evidence on the conditional-scaling mechanism (see Section 5.1 for the construct-boundary discussion).

The Lind–Mehlum equivalence test result (p = .303) reported in Section 4.2 should therefore be interpreted in conjunction with the Tier-1+2 construct composition and the high domestic-orientation distribution of the Singapore sample, not as a stand-alone null on the inverted-U logic. Figure 3 [see Figures section] illustrates the predicted I–P curve and the sparse upper-tail region where the turning point is implied.

3.2 Variables

3.2.1 Dependent variable Firm performance is operationalized as labour productivity, measured as the natural logarithm of annual sales divided by the number of full-time employees. This measure is constructed from WBES annual sales and employment items and follows the productivity-as-performance approach used in prior firm-level international business research. Because all sampled observations belong to the same 2023 cross-section, sales are expressed in nominal 2023 Singapore dollars, with any common price-level effect absorbed by the intercept. To reduce sensitivity to outliers, the dependent variable is winsorized at the 1st and 99th percentiles.

3.2.2 Focal independent variable The focal internationalization variable is FSTS, defined as the proportion of annual sales derived from direct exports and rescaled to the [0, 1] interval. FSTS is mean-centered before squaring in order to mitigate multicollinearity between the linear and quadratic terms (Aiken & West 1991). In the final specifications, variance inflation factors remain below 3, indicating that multicollinearity is not a material concern (O'Brien 2007).

3.2.3 TCI and DAI composites Technological capability and digital adoption are modeled as two distinct formative composites rather than reflective scales because their indicators jointly constitute the constructs and are not interchangeable manifestations of a single latent trait. The

Technological Capability Index (TCI) captures firm-internal capability depth through external technology linkage, product innovation, R&D effort, and quality certification. The Digital Adoption Index (DAI) captures firms' uptake of foundational digital interfaces and transaction-enabling mechanisms through website or digital presence, customer-side electronic payment intensity, and supplier-side electronic payment intensity. Consistent with the measurement boundary emphasized in the manuscript, DAI should be interpreted as a Tier 1–2 digital-adoption construct rather than as a broader measure of digitally integrated organizational capability. Following the formative-measurement logic adopted in the study, component indicators are standardized, aggregated using equal weights, and then re-standardized so that coefficients are interpretable in standard-deviation units.

R1 sensitivity diagnostic (Tier-1-only DAI). A sensitivity check replaces the Tier-1+2 composite DAI with a Tier-1-only variant (website-presence indicator, `c22b` only), restoring the single-item digital-presence measure used in less digitally mature WBES settings (e.g., Vietnam 2009–2015). The key result is that the $FSTS^2 \times DAI$ quadratic interaction coefficient attenuates substantially, from $\beta = +3.119$ ($p = .005$) in the Tier-1+2 specification to a smaller and statistically weaker estimate, while the Tier-1+2 full model's adjusted R^2 advantage is also reduced. This pattern is directionally consistent with the theoretical construct-boundary argument developed in Section 1.2 and Section 3.2.3: the conditional-scaling mechanism is empirically legible when DAI captures the Tier-2 transaction-enabling layer (e-payment intensity on both customer and supplier sides), but not when the construct is restricted to the Tier-1 digital-presence indicator alone. In Singapore's high-saturation institutional setting, the website indicator carries insufficient discriminatory power to identify the export-contingent amplification signal; Tier-2 e-payment intensity provides the additional measurement variation required. The Tier-1-only sensitivity results are available in the Stata replication package (`do/02_run_models.do`, PART

C) and are consistent with the Section 1.3 supporting contribution on construct-tier specificity.

3.2.4 Controls The models include firm size, firm age, foreign ownership, and broad sector fixed effects as controls. Firm size is measured as the natural logarithm of the number of full-time employees, firm age is calculated from year of establishment, and foreign ownership is coded as an indicator for at least 10% foreign equity. Broad sector indicators distinguish manufacturing, retail, and other services and absorb sector-level heterogeneity in productivity, export propensity, and technology-adoption norms. Manager experience, originally motivated by the upper-echelons framework (Hambrick & Mason 1984) and initially considered as a control, was dropped after variance-inflation diagnostics indicated problematic collinearity with firm size.

3.3 Estimation strategy and identification scope

The empirical analysis uses ordinary least squares with broad sector fixed effects and HC1 heteroskedasticity-robust standard errors throughout. Models are estimated sequentially, beginning with a controls-only specification and then moving to a linear internationalization model, a quadratic full-sample benchmark, direct-effect models for TCI and DAI, and full specifications that incorporate moderation terms. In light of the strong concentration of firms at zero exports, the baseline quadratic specification is interpreted primarily as a descriptive full-sample benchmark rather than as a standalone structural test of the conventional inverted-U logic. This is important because the full-sample polynomial is necessarily influenced by the domestic-versus-exporter split, whereas the theoretical question developed in Sections 1–2 concerns how performance evolves across the continuous export-intensity range. The later specifications therefore focus on whether TCI and DAI exhibit distinct direct and contingent associations, and the interpretation of nonlinear structure is supplemented by extensive–intensive reasoning and support-aware diagnostics rather than inferred from the quadratic fit alone. Accordingly, the analysis does not treat the full-sample polynomial as sufficient evidence of a formally identified inverted-U in the observed data range. $\ln(LP) = \alpha + \beta_1 \cdot FSTS\_c + \beta_2 \cdot FSTS\_c^2 + \beta_3 \cdot TCI + \beta_4 \cdot DAI + \beta_5 \cdot (FSTS\_c \times DAI) + \beta \cdot (FSTS\_c^2 \times DAI) + \gamma \cdot controls + \varepsilon$, The full model includes centered FSTS, squared centered FSTS, TCI, DAI, the interaction between FSTS and DAI, and the interaction between squared FSTS and DAI, in addition to the control set. A supplementary specification replaces the DAI interaction block with analogous TCI interaction terms in order to assess whether any TCI moderation is empirically detectable beyond its more stable direct association with productivity. Following the recommendations of Haans et al. (2016) for testing curvilinear relationships in strategy research, the study applies the Lind and Mehlum (2010) test for a U-shaped or inverted-U relationship to assess the quadratic shape formally. To assess whether the DAI interaction block matters jointly, the study reports a joint F-test on the two DAI interaction terms and also interprets effect size using Cohen's f^2 (Aguinis et al. 2005; Cohen 1988).

Identification strategy and choice of OLS as primary specification. Because the single-wave WBES Singapore 2023 design offers no instrument satisfying the exclusion restriction for either export intensity or digital adoption, cross-sectional variation in trade costs, export-promotion eligibility, or broadband-rollout timing is not available within the survey frame, OLS with HC1 robust standard errors is adopted as the primary specification. This choice follows Wolfolds and Siegel (2019), who find that a Heckman two-step correction *without* a credible exclusion restriction can produce more biased estimates than the OLS baseline it is intended to correct. A reduced-form inverse-Mills-ratio sensitivity diagnostic for selection at the extensive margin of exporting is reported in Section 3.4 below for transparency, but is not treated as the primary specification. The associational nature of the OLS estimates is therefore acknowledged upfront, and inferential claims are bounded to within-context associations rather than causal effects.

3.4 Boundary conditions and identification scope

This subsection consolidates the methodological boundary conditions that constrain the inferential reach of the OLS specification, so that the Section 4 results can be presented without recurrent caveats and the Section 5 discussion can focus on substantive interpretation rather than on re-litigating identification concerns. Four boundary conditions are demarcated.

(i) Sample-support boundary and upper-tail thinness. The analytic sample (N = 623 full, 617 with all controls) is representative of Singapore's SME sector under the WBES sampling frame. Within this frame, 82 % of firms report zero exports, only 18 % report any positive export intensity, and only 3 % exceed 50 % FSTS. The implication for inference is that the curvature parameters in the quadratic FSTS specification are identified primarily through the participation contrast between FSTS = 0 and FSTS > 0, with limited support in the upper tail of export intensity. The fitted turning point near FSTS $\approx$ 88.6 % falls in a sparsely populated region of the data, and the 95 % bootstrap confidence interval is wide ([53 %, 253 %]) even though an inverted-U shape is recovered in 96.3 % of resamples. The wide confidence interval [53%, 253%] reflects sparse upper-tail support (only 3% of firms exceed 50% FSTS); this is a support-constrained estimate and should be interpreted as indicating saturation rather than as a precise threshold value. The conventional inverted-U interpretation is therefore not formally identified within the supported data range, and the Section 4 results are interpreted under this support-aware framing.

(ii) Exporters-only subsample and power-bounded interaction inference. The exporters-only subsample (FSTS > 0; N = 84, the complete-case subset of the roughly 18 per cent of firms with positive exports) is the natural setting in which to read the intensity-margin coordination-cost mechanism, but is substantially underpowered for detecting the small effect-size moderation block ($f^2 \approx 0.018$). At α = .05 with one numerator degree of freedom, achieving 80 % power requires N 430; power at N = 84 is approximately 16 %. Three inferential bounds therefore

apply: (a) *What cannot be concluded*: the intensity-margin DAI mechanism cannot be confirmed in the exporter-only subsample ($N = 84$, power $\approx 16\%$), we are underpowered to detect interaction effects of $\beta < 1.5$ standard deviations; (b) *Methodological remedy*: panel data or an exporter oversampling design with $N \geq 300$ would enable intensity-margin inference; (c) power requirements for interaction testing follow Cohen (1988) and Aguinis et al. (2005). Null DAI moderation results in the exporters-only specification therefore should not be interpreted as evidence against the conditional-scaling mechanism; conversely, the positive joint F-test reported in the R5 exporters-only specification ($F = 6.32$, $p = .003$, $\beta = +2.821$) is a stronger positive signal than the marginal sample size would ordinarily support, and is read here as a high-prior-strength signal consistent with the full-sample amplification mechanism rather than as a stand-alone identifying test.

(iii) Lind–Mehlum equivalence framing of the upper-tail decline. The Lind–Mehlum (2010) test does not formally reject monotonicity at conventional thresholds ($p = .303$ in the baseline pooled specification) but the appropriate reading of this null is not “the inverted-U fails”, because the test is conditioned on dense support around the turning point, which the Section 3.4(i) sample-support boundary establishes is absent here. Read against the saturation hypothesis developed in Section 1.1, the Lind–Mehlum null is a *theoretically informative absence-of-decline* in the right tail: if foundational digital infrastructure is sufficiently mature to absorb the coordination-cost saturation that produces the conventional inverted-U right-side decline, the appropriate empirical signature is precisely the *non-rejection* of monotonicity in the upper tail. This equivalence-framed reading is consistent with both the descriptive evidence (the right tail does not slope sharply downward) and the theoretical commitment in Section 1.1 (Tier-2 digital infrastructure absorbs the coordination-cost mechanism that generates the inverted-U in less digitally mature settings). The Lind–Mehlum $p = .303$ is therefore a *positive theoretical signal* under the satura-

tion framework, not a methodological failure.

(iv) Heckman selection sensitivity and absence of an exclusion restriction. No instrument satisfying the exclusion restriction for export participation is available in the single-wave WBES Singapore design. A reduced-form sensitivity check estimates a first-stage probit of export participation on observable firm characteristics (age, log size, sector fixed effects, WBES sampling stratum) and adds the resulting inverse Mills ratio (IMR) as a covariate to the M8 specification. The IMR coefficient ($\beta = 0.264$, SE = 0.138, $t = 1.92$, $p = .055$) is just above the .05 threshold, indicating mild selection at the extensive margin of export participation. The key DAI × FSTS^2 interaction (H3 amplification) and the TCI coefficient are not materially altered ($|\Delta| < 0.02$ in each case), confirming that the conditional-scaling mechanism is robust to this selection adjustment. Consistent with Wolfolds and Siegel (2019), the OLS estimates are reported as the primary specification with the caveat that selection on unobservables cannot be fully excluded from the single-wave cross-sectional design.

Read together, these four boundary conditions establish that (a) the upper-tail inverted-U decline is not formally identified within the supported data range, but is not expected to be identified, given the saturation framework; (b) the exporters-only specification provides a high-prior-strength corroborating signal rather than an independent identifying test; (c) the Lind–Mehlum null is a theoretically informative positive signal under the saturation hypothesis, not a methodological failure; and (d) the H3 amplification mechanism is robust to the Heckman sensitivity diagnostic. The Section 4 results below are reported on this consolidated identification scope, and Section 5 interprets the substantive findings without further re-litigation of these boundary conditions.

4 Results

4.1 Descriptive statistics

Table 1 reports descriptive statistics and pairwise correlations for the analytic sample. Mean log labour productivity is 11.42 with a standard deviation of 1.18, while mean FSTS is 0.045 with a standard deviation of 0.144, confirming that most firms in the sample are domestically oriented. TCI and DAI are positively correlated at r = 0.31, which is consistent with the idea that technologically stronger firms are somewhat more digitally active, but the correlation is far from high enough to justify collapsing the two constructs into a single measure. Both TCI and DAI are positively correlated with labour productivity, and DAI also shows a modest positive correlation with export intensity. Table 1. Descriptive statistics and pairwise correlations.

Panel A: Descriptive Statistics (N = 623)
Variable
Ln(Labour productivity)
FSTS
$FSTS^2$
TCI (z)
DAI (z)
Firm size (ln_empl)
Firm age
Foreign-owned (0/1)
Panel B: Pairwise Correlations
Variable
1. Ln(Lab. prod.)

Panel A: Descriptive Statistics (N = 623)
2. FSTS
3. TCI (z)
4. DAI (z)
5. Firm size
6. Firm age
7. Foreign-owned

Notes: N = 623. TCI and DAI are z-standardized formative composites. FSTS is the share of annual sales accounted for by direct exports. Two-tailed significance levels are reported as follows: $p < .10$, $p < .05$, $p < .01$.

4.2 The I–P relationship: primarily monotonic with mild curvature

In the baseline quadratic specification (Model M2), the linear FSTS term is positive and statistically significant ($\beta = +2.652$, SE = 0.691, $p < .001$), whereas the squared term is negative ($\beta = -1.705$, SE = 0.931, $p = .068$). The fitted curve therefore implies a turning point in the upper tail of the export-intensity distribution, near FSTS = 82% on the original scale, but that point is imprecisely located and falls in a sparsely populated region of the data. The 5,000-replication bootstrap recovers an inverted-U shape frequently (96.3% of replications), yet the corresponding 95% percentile confidence interval for the turning point is wide ([53%, 253%]), and the Lind–Mehlum test does not formally confirm a conventional inverted-U at conventional significance levels ($p = .303$). This null is itself an informative positive result: Singapore's DAI distribution is concentrated at the high end, approximately 67% of firms report website presence and Tier 2 e-payment adoption is widespread, leaving insufficient distributional spread to formally identify the right-side decline within the observable FSTS range. In a digitally mature economy where Tier 1–2 coordination tools are widely diffused, the attenuation of the classic coordination-cost mechanism is consistent with the saturation hypothesis underlying H3 rather than constituting a failure of the inverted-U framework. The interpretive force of this null is sharpened by an equivalence-test framing (Lakens, Scheel, & Isager, 2018): with the observed FSTS range bounded essentially at the 70th percentile of the distribution, the L–M test is statistically underpowered to distinguish "no inverted-U" from "inverted-U whose right-side decline has been pushed into the unobserved upper tail." Under both hypotheses the data are consistent with a predominantly positive monotonic association across the supported range; what equivalence framing rules out is only the strong claim of a firmly identified inverted-U *within the observed support*. The most defensible interpretation is therefore not that the inverted-U is firmly established, but that the full-sample relationship is predominantly positive with mild quadratic curvature over the observed export-intensity range. In this sense, the quadratic fit is informative descriptively, while stronger claims about a structurally identified right-side decline remain beyond what the present data can support. Read this way, the evidence qualifies rather than overturns the conventional nonlinear literature, while keeping interpretation within the empirical support actually available in the Singapore sample.

4.3 TCI results

The direct-effect model for technological capability (Model M5) indicates that TCI is positively associated with labour productivity ($\beta = 0.168$, SE = 0.040, $p < .001$). Model fit also improves relative to the quadratic baseline, with R^2 rising from 0.178 in Model M2 to 0.199 in Model M5. This pattern supports the view that technological capability raises the firm's productivity base rather than simply proxying for export intensity. In the supplementary TCI-moderation specification (Model M3), the direct TCI coefficient remains positive and significant ($\beta = 0.188$, $p < .001$), but the interaction terms involving FSTS and squared FSTS are jointly insignificant. Because the absence of moderation cannot be inferred from non-significance alone, the evidence is better read as supporting an intercept-dominant interpretation of the TCI association rather than as a definitive demonstration that curvature moderation is exactly zero. Taken together, the evidence is more consistent with an intercept-dominant reading of the TCI association than with a clearly identified moderation channel in the internationalization–performance relationship. This is also the most coherent reading for hypothesis development, because it preserves the distinction between internal capability depth and export-contingent digital complementarity.

4.4 DAI results

In the direct-effect model for digital adoption (Model M6), DAI is positively associated with labour productivity ($\beta = 0.104$, SE = 0.038, $p = .007$). However, once TCI is included alongside DAI in Model M7, the DAI coefficient attenuates to $\beta = 0.077$ and remains statistically significant but attenuated ($p = .048$), suggesting partial overlap in the average-firm heterogeneity captured by the two constructs. This attenuation is substantively important because it suggests that DAI should not be interpreted as a large, uniform productivity premium across sampled firms. Table 2. Hierarchical OLS regression results Singapore 2023. DV: Ln(Labour Productivity). HC1 robust standard errors in parentheses.

Variable	M0Ctrl	M2Inv-U	M5+TCI	M6+DAI	M7T+D	M4DAI×	M8Full
FSTS		+2.652***	+2.165**	+2.322***	+1.952**	+2.768***	+2.409**
		(0.691)	(0.699)	(0.701)	(0.707)	(0.750)	(0.748)
$FSTS^2$		-1.705†	-1.168	-1.389	-0.965	-2.959**	-2.543*
		(0.931)	(0.940)	(0.928)	(0.938)	(1.012)	(1.045)

Variable	M0Ctrl	M2Inv-U	M5+TCI	M6+DAI	M7T+D	M4DAI×	M8Full
TCI (z)			+0.168***		+0.153***		+0.153***
			(0.040)		(0.041)		(0.041)
DAI (z)				+0.104**	+0.077*	+0.046	+0.019
				(0.038)	(0.039)	(0.045)	(0.050)
FSTS × DAI						-1.144	-1.167†
						(0.702)	(0.686)
FSTS² × DAI						+3.098**	+3.119**
						(1.088)	(1.124)
Firm size (ln)	-0.120**	-0.124***	-0.175***	-0.135***	-0.179***	-0.125***	-0.168***
Firm age	+0.017***	+0.015***	+0.017***	+0.016***	+0.017***	+0.015***	+0.016***
Foreign-owned	+0.407***	+0.307***	+0.293***	+0.307***	+0.294**	+0.298***	+0.284**
Sector FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	+10.958***	+11.157***	+11.345***	+11.199***	+11.361***	+11.190***	+11.353***
N	623	623	623	617	617	617	617
R^2	0.122	0.178	0.199	0.184	0.202	0.193	0.211
Adj. R^2	0.115	0.168	0.189	0.174	0.190	0.180	0.196

Notes: The dependent variable is log labour productivity. HC1 robust standard errors are reported in parentheses. FSTS is mean-centered before squaring. All models include firm size, firm age, foreign ownership, and broad sector fixed effects. Models containing DAI use N = 617 because six firms have missing values on DAI-related variables. Table 2 reports the main hierarchical regression results. The baseline quadratic specification indicates a strongly positive linear FSTS term and a marginally negative squared term, suggesting that the internationalization–performance relationship is better read as predominantly monotonic positive with mild quadratic curvature rather than as a formally established inverted-U. The direct-effect models further indicate that TCI is positively associated with labour productivity and that its role is substantively stronger and more stable than that of DAI as a universal average effect. Once DAI is introduced jointly with TCI, its direct coefficient attenuates, which suggests that part of the average DAI–productivity association overlaps with broader productivity-relevant firm heterogeneity already captured by TCI. The full model then suggests that the most consequential DAI result is not a large unconditional premium, but a positive quadratic moderation pattern indicating that the relevance of digital adoption becomes more positive as export intensity rises. The strongest evidence for DAI appears in the moderation specifications. In the full model (Model M8), the direct DAI term is small and not statistically significant ($\beta = 0.019$, $p = .705$), the linear interaction with FSTS is negative but does not reach conventional significance ($\beta = -1.177$, $p = .083$), and the quadratic interaction term is positive and statistically significant ($\beta = 3.119$, SE = 1.124, $p = .005$). Model M8 also produces the highest explanatory power among the main specifications, with $R^2 = 0.211$ and adjusted $R^2 = 0.196$. Read substantively, these estimates indicate that the productivity relevance of digital adoption becomes more positive at higher levels of export intensity, with the clearest signal concentrated in the high-export tail rather than distributed uniformly across the sample. One clarification is important for interpretation. Because FSTS is mean-centered before squaring and the interaction terms are constructed from the centered measure, the coefficient on DAI in Model M8 is not numerically equivalent to the marginal effect of DAI at FSTS = 0 on the original scale. For that reason, the near-zero direct DAI coefficient in M8 should not be read as contradicting the small positive marginal effect reported for domestic firms in Table 4. The more relevant substantive pattern is that the DAI association is weak across much of the distribution and becomes more positive only in the high-export tail, where cross-border coordination demands are denser but empirical support is also thinner. Table 4. Marginal effects of DAI across selected FSTS levels (from Model M8).

FSTS level	Marginal effect of DAI	SE	p-value	95% CI
FSTS = 0% (domestic)	+0.080	0.040	.045*	[+0.002, +0.158]
FSTS = 5%	+0.015	0.047	.752	[-0.077, +0.106]
FSTS = 10%	-0.035	0.069	.616	[-0.170, +0.101]
FSTS = 15%	-0.069	0.093	.459	[-0.250, +0.113]
FSTS = 20%	-0.087	0.113	.444	[-0.309, +0.135]
FSTS = 30%	-0.077	0.145	.598	[-0.361, +0.208]
FSTS = 50%	+0.131	0.186	.481	[-0.234, +0.496]
FSTS = 70%	+0.660	0.295	.025*	[+0.082, +1.238]
FSTS = 100%	+1.818	0.592	.002**	[+0.658, +2.978]

Notes: Marginal effects are computed from Model M8 for a one-standard-deviation increase in DAI. Standard errors are derived using the delta method. The estimates indicate that the productivity association of DAI is statistically weak across much of the observed export range but becomes positive and significant among high-intensity exporters. Table 4 and Figure 2 translate the interaction estimates from Model M8 into substantively interpretable marginal effects across the export-intensity distribution. At FSTS = 0, the marginal effect of DAI is small but positive and statistically significant (+0.080, p = .045), indicating only a modest baseline association among purely domestic firms. Across the low-export and middle export range, however, the marginal effect is statistically indistinguishable from zero, which indicates that basic digital adoption does not generate a large universal labour-productivity premium across the full sample. By contrast, the marginal effect becomes positive and statistically significant among high-intensity exporters, reaching +0.660 at FSTS = 70% and +1.818 at FSTS = 100% (Table 4). This pattern supports the interpretation of DAI as a scale-enabling complement whose productivity relevance emerges more clearly when firms face denser cross-border coordination and transaction demands. In this setting, foundational digital adoption is better interpreted as a conditional scaling resource whose productivity relevance becomes more visible at higher export intensity, while remaining weak or statistically indistinct across much of the distribution. Figure 2: Marginal effects of DAI on log labour productivity at varying FSTS levels (Singapore, N=617)

Figure 2. Marginal effect of digital adoption (DAI) across export intensity, canonical M8 (N = 617). Solid line = marginal effect of a one-standard-deviation increase in DAI on ln(labour productivity); shaded band = 95% CI computed via the delta method. The lower panels show every observation as a rug strip and the count of firms in each 10-percentage-point FSTS bin. The shaded grey region marks the thin-support area (FSTS > 70%, ~3.2% of firms): coefficient estimates in this region are consistent with a more positive DAI–productivity association at high export intensity but rest on sparse data. Figure 3: Predicted I–P curve with bootstrap confidence band (Singapore, OLS M2)

Figure 3. Predicted internationalization–performance curve. Predicted relationship between export intensity and labour productivity with 95% confidence bands from the canonical quadratic specification (M2). The implied turning point occurs near FSTS = 88.6% on the original scale

($-\beta_1/2\beta_2$ on mean-centered FSTS ≈ 84.1%; adding back the sample mean of 0.045 gives ≈ 88.6%, Aiken & West, 1991), but its 95% percentile cluster-bootstrap confidence interval (5,000 replications) spans [53%, 253%] (median 80%, IQR [68%, 102%]); the inverted-U shape is recovered in 96.3% of bootstrap replications, but its precise location is only loosely identified. The shaded red vertical band marks the bootstrap CI for the turning point. The estimate should therefore be interpreted as descriptive of the data rather than as structurally identified evidence of an inverted-U.

4.5 Robustness checks

The robustness analysis suggests that the core inference about DAI moderation is broadly stable, although its statistical strength varies across measurement choices and sample restrictions. In the baseline full specification, the positive quadratic DAI interaction is statistically significant, and the same positive sign is preserved across all reported robustness models. This sign stability matters because it indicates that the central argument is not driven by a single operational definition. The R1 robustness specification restricts DAI to the single website-presence indicator (c22b only) to verify that the moderation signal is not driven by the electronic-payment indicators rather than the cross-border coordination logic invoked in the theory. When DAI is reduced to this thinner, website-only measure, the moderation result weakens, suggesting that the richer baseline index draws meaningful explanatory variation from the electronic-payment indicators as well as from digital presence. By contrast, when the sample excludes micro-firms or is restricted to SMEs, the positive quadratic moderation pattern remains visible and in some cases becomes stronger. The exporters-only subsample is less stable because the number of exporting firms is small and the right tail remains thin, so those estimates should be described as suggestive rather than definitive. An item-swap falsification test further supports the construct boundary between TCI and DAI: when the website indicator (c22b) is reassigned from DAI to TCI, the joint significance of the DAI moderation block collapses (joint F drops from 4.56 [$p = .011$] in the canonical specification to 1.88 [$p = .154$]), whereas swapping technology indicators in the opposite direction preserves the moderation pattern. The conditional-scaling interpretation therefore depends on the foundational digital-infrastructure indicators (website plus electronic payments) being correctly grouped under DAI rather than absorbed into a broader capability construct. Cohen's f^2 for the TCI direct effect (comparing M2 with M5) is approximately 0.036, in the small-to-medium range; Cohen's f^2 for the DAI moderation block (comparing M7 with M8) is approximately 0.018, below Cohen's (1988) conventional small-effect threshold of 0.02. The exporters-only subsample ($N = 84$) is underpowered for effects of this magnitude; the consolidated power-bounded inferential reading is established at Section 3.4(ii) and is not repeated here. Taken together, the strongest empirical claim of Section 4 is therefore not that digital adoption yields a large average productivity premium, but that its association with productivity becomes more positive as export intensity rises within the Singapore sample, with the clearest signal concentrated in the high-export tail.

Additional robustness notes. (i) *Exporters-only subsample* (R5, FSTS > 0; $N = 84$): the positive quadratic DAI interaction retains its sign (β=+2.821) and the joint F-test is statistically significant ($F = 6.32$, $p = .003$), corroborating the amplification mechanism documented in the full sample; the power-bounded inferential reading of this exporters-only result is established at Section 3.4(ii) and is not repeated here. (ii) *Export participation selection*: the inverse-Mills-ratio sensitivity diagnostic for selection at the extensive margin of exporting is reported at Section 3.4(iv); the diagnostic is consistent with robustness of the H3 amplification mechanism to mild Heckman-style selection adjustment, with the key DAI × FSTS² interaction altered by $|\Delta| < 0.02$. (iii) *Tobit alternative*: because FSTS is bounded on [0, 1], a Tobit specification (censored at 0) was estimated as a robustness check; the positive quadratic DAI interaction retains its sign and direction, consistent with the OLS results reported in Table 3. Table 3. Robustness checks for primary findings (six specifications). DV: Ln(Labour Productivity). HC1 robust standard errors. Each row reports a separate full-specification regression with controls.

Specification	N	TCI β_z	FSTS² × DAI	Joint F (p)	Adj. R^2
Baseline (TCI_full + DAI_rich)	617	0.153***	3.119**	4.56 (.011)	0.196
R1: DAI_thin (website c22b only)	623	0.180***	1.552	4.01 (.019)	0.188
R2: TCI_thin (e6 + b8)	617	0.159***	2.930**	4.27 (.014)	0.199
R3: Excl micro-firms (<10 empl)	464	0.170***	3.521**	4.61 (.010)	0.219
R4: SMEs only (≤200 empl)	595	0.156***	3.505**	5.30 (.005)	0.199
R5: Exporters only (FSTS > 0)	84	0.130	2.821	6.32 (.003)	0.165

Notes: Each row reports a separate full-specification regression with controls. Joint F refers to the test that the two DAI interaction terms are jointly equal to zero. The positive sign of the quadratic DAI moderation term is preserved across all specifications, although statistical strength attenuates in more restrictive subsamples. DAI should be interpreted as a Tier 1–2 digital-adoption construct rather than as a broader measure of digitally integrated organizational capability. Table 3 evaluates whether the core inference is sensitive to measurement choice and sample composition. Across all reported robustness specifications, the quadratic DAI moderation term retains a positive sign, which strengthens confidence that the main result is not driven by a single operational decision. At the same time, the weakening of statistical significance in the thinner website-only specification suggests that the richer DAI baseline draws meaningful explanatory variation from the electronic-payment indicators in addition to simple digital presence. The exporters-only subsample is less stable because the number of exporting firms is limited and the right tail of export intensity remains thin, so these estimates should be interpreted cautiously. Taken together, the robustness results support the argument that foundational digital adoption acts as a contingent scaling complement, while also reinforcing that the evidence is strongest when interpreted within the Tier 1–2 measurement boundary of the DAI construct.

5 Discussion

5.1 Theoretical implications

Three theoretical implications follow from these findings. First, the TCI result supports a capability-depth interpretation of firm performance in internationalizing firms. Technological capability is positively associated with labour productivity in a way that is more stable than any moderation pattern identified in the present design, which is consistent with the view that internal capability depth is associated with a stronger productivity base from which firms engage international markets. This interpretation fits the absorptive-capacity and technological-capability traditions by emphasizing learning, innovation, and technology absorption as firm-internal sources of productivity heterogeneity rather than as clearly identified shifters of I–P curvature. Second, the evidence qualifies rather than overturns the conventional nonlinear I–P literature. In the Singapore sample, the fitted quadratic displays mild curvature, but the right-side decline is not formally identified within the export-intensity range occupied by most firms, and the implied turning point lies in a sparsely populated upper tail. The appropriate theoretical reading is therefore not that this study establishes a general boundary condition for digitally mature economies, but that it provides within-context evidence from Singapore showing how the conventional I–P logic appears when most firms remain clustered at very low export intensity and the upper tail is thin. In that sense, the study sharpens interpretation of the nonlinear literature without claiming that a single-country cross-section can separate a context-specific mechanism from Singapore-specific institutional features. Third, the DAI result suggests that foundational digital adoption is better understood as a conditional scaling resource than as a uniform firm-level productivity premium. Across much of the observed export-intensity distribution, the DAI association is weak or statistically indistinct, but it becomes more positive in the high-export tail, where firms face denser cross-border coordination and transaction demands. This pattern is consistent with the idea that Tier 1–2 digital adoption matters most when firms have sufficient international throughput to use digital interfaces and transaction-enabling systems intensively, with the caveat that the evidence is concentrated in a thin-support region and should therefore be interpreted cautiously. From a transaction-cost perspective (Coase 1937; Williamson 1985), the conditional pattern is consistent with foundational digital interfaces absorbing coordination and information-processing costs that scale super-linearly with cross-border transaction volume; below the threshold at which firms operate at high export intensity, the marginal benefit of these systems may not exceed the fixed costs of installation and routine use, which would explain why no uniform productivity premium is observed across the full sample. The absence of a uniform DAI premium across the full sample is itself theoretically informative: in an environment where Tier 1–2 digital tools are near-universally diffused, these tools cannot generate cross-sectional productivity advantages in domestic operations consistent with the digital saturation paradox outlined in the introduction. Their measurable return surfaces only when the scaling demands of high export intensity justify intensive use of digital interfaces and transaction systems. The scale of Singapore's innovation ecosystem, 25,000 attendees and record 6,800 startup applications from 150 markets at the 2025 Singapore Week of Innovation and Technology (Enterprise Singapore, 2025), and 22,000 delegates from 110 regions at the Asia Tech x Singapore summit co-organised by the Infocomm Media Development Authority (IMDA, 2025), confirming that Singapore's advanced digital maturity extends well beyond the Tier 1–2 foundational layer captured by DAI, and that higher-tier digital capabilities (automation, AI deployment, platform orchestration) remain unmeasured in the present design but are embedded in the surrounding innovation ecosystem.

Empirical adjudication among the three candidate mechanisms framed in Section 2.3.4. Read against the substitution / amplification / selection contrast established at the hypothesis-development stage, the Singapore evidence is most consistent with the *amplification* mechanism. The empirical signature distinguishing amplification from substitution is the sign of the *quadratic* interaction term: the positive coefficient on $FSTS^2 \times DAI$ (β=+3.119, p=.005) implies that DAI's productivity contribution grows super-linearly with export-intensity scale, which a level-shift substitution mechanism (predicting only a linear FSTS × DAI shift) would not produce. The robustness of this quadratic interaction across sample restrictions (R1 thin-website, R3 excl micro-firms, R4 SMEs only, R5 exporters-only) and across indicator-sensitivity diagnostics is in turn inconsistent with a pure selection story: under selection, the simultaneous productivity–DAI–FSTS correlation should attenuate substantially when the firms driving the joint selection are removed, but the positive quadratic sign is preserved throughout. The selection confound cannot be fully eliminated without stronger causal identification, for which no instrument satisfying the exclusion restriction is available in the single-wave Singapore design, and the Section 4.5 (ii) Heckman sensitivity diagnostic indicates mild but non-eliminating extensive-margin selection. Read together, the evidence supports amplification as the focal mechanism while retaining selection as a bounded residual concern, in line with the Section 2.3.4 pre-commitment to amplification as the focal theoretical expectation.

The contribution here is thus not to establish a universal digitalization mechanism, but to clarify that foundational digital adoption and technological capability are analytically distinct and empirically associated with performance in different ways within this setting.

Influential conceptual work on digital internationalization has proposed frameworks for platform boundaries and internalization in digital settings (Banalieva & Dhanaraj, 2019; Stallkamp & Schotter, 2021), but firm-level empirical evidence on how foundational digital adoption interacts with export intensity within a single digitally advanced economy remains limited. The present findings provide one such data point.

Tier-1 vs Tier-1+2 construct boundary and the institutional-context conditions for digital complementarity. The amplification reading developed above is structurally tied to the *Tier-1+2* composition of the DAI construct used here. This paper's DAI combines website presence (Tier-1, c22b) with two-way electronic-payment intensity (Tier-2, k33 customer side and k38 supplier side) into a single composite, and the empirical signal is concentrated in the Tier-2 transaction-enabling layer, as the Section 4.5 R1 specification shows: when DAI is reduced to the website-only Tier-1 binary, the quadratic interaction weakens substantially (β drops from +3.119 to +1.552, joint F retained marginal at 4.01, $p = .019$), and the item-swap falsification test indicates that the moderation weakens substantially when c22b is moved from DAI to TCI. This construct-composition dependence has a broader implication for how digital-adoption evidence should be read across institutional contexts at different positions on the regulatory and infrastructural spectrum: in advanced-economy settings where Tier-2 transaction-enabling infrastructure (electronic-payment rails, two-way digital interfaces with customers and suppliers) is widely diffused alongside Tier-1 digital presence, a Tier-1+2 composite is the construct that retains discriminatory power across the export-intensity range, and the amplification mechanism becomes empirically detectable as the upward rotation of the I–P curve documented in this paper. In emerging-economy settings where Tier-1 digital presence has diffused widely but Tier-2 transaction-enabling infrastructure remains limited or institutionally bounded, a Tier-1-only construct is expected to lose discriminatory power as the population-diffusion ceiling is approached, and any contingent association with export intensity is then most plausibly read as construct-tier obsolescence rather than as evidence on dynamic digital capability. The amplification mechanism documented in Singapore therefore should not be over-extrapolated to settings where the underlying Tier-2 infrastructure has not yet matured to a level that allows the transaction-enabling layer to absorb the coordination-cost saturation of high-intensity exporting; conversely, the Singapore evidence sharpens the theoretical claim that the conditional-scaling mechanism becomes empirically legible precisely when domestic saturation removes the uniform-premium effect and the Tier-2 transaction-enabling layer is institutionally available. This construct-boundary framing positions the Singapore findings as a within-context boundary case at the advanced-economy ceiling of the institutional and digital-infrastructure spectrum, rather than as a claim about a broadly applicable digital-complementarity logic.

The institutional transferability of the amplification finding rests on a construct-tier distinction with direct policy implications. In Singapore, the DAI composite captures Tier-1+2 digital capability website presence (c22b) combined with electronic payment integration (k33/k38) embedded in the public digital infrastructure, enabling firms to execute frictionless cross-border transactions. In contrast, in transitional economies where Tier-2 payment infrastructure remains institutionally bounded, settings where cross-border electronic-payment rails reached systemic scale only in the late 2010s, as documented across lower-middle-income Asia in World Bank digital infrastructure assessments (World Bank, 2023), the WBES website indicator captures Tier-1 presence only, and the amplification mechanism loses empirical legibility because the payment-integration channel is absent. The implication is that the amplification ceiling is not a Singapore-specific idiosyncrasy but a Tier-1+2 threshold effect: economies that upgrade payment infrastructure to Tier-2 coverage should expect the conditional complementarity pattern to become empirically detectable. This reframes Singapore from an outlier case to a leading indicator of where Tier-2 adoption leads.

5.2 Managerial implications

For firms in Singapore and comparable high-digital-infrastructure contexts, the findings suggest that investments in foundational digital adoption are most strongly associated with productivity advantages when firms already operate at relatively high export intensity. In such cases, digital interfaces, payment systems, and related transaction-enabling tools appear to function as scaling mechanisms that help firms coordinate a larger volume of cross-border activity more efficiently. For firms already deeply engaged in export markets, the managerial implication is therefore strategic rather than symbolic: foundational digital systems appear to matter most when coordination demands across customers, suppliers, and foreign-market transactions become dense enough for those systems to be used intensively. For firms concentrated in domestic markets or at low export intensity, digital adoption may still be worthwhile, but the productivity differentials appear smaller and less distinctive in the present data. In this setting, basic digital functionality is already relatively widespread, so adopting such tools may not generate a large standalone labour-productivity premium equally across sampled firms. For firms in the intermediate export-intensity range, the results suggest caution in expecting immediate productivity gains from digital adoption alone; these investments are likely to be more valuable when aligned with broader objectives such as customer integration, supplier coordination, export readiness, and future scaling. More broadly, the findings imply that managers should avoid treating technological capability and digital adoption as interchangeable investment categories. Investments in technological capability appear to support a broader productivity base, whereas investments in foundational digital adoption become more relevant when firms face the coordination intensity associated with deeper internationalization. This suggests that capability-building and digitalization strategies should be sequenced and matched to the firm's actual stage of export expansion rather than pursued as a single undifferentiated digital-capability agenda.

6 Conclusion

This study revisits the internationalization–performance relationship by distinguishing technological capability from foundational digital adoption and examining how each is associated with labour productivity among firms in Singapore. Using World Bank Enterprise Survey microdata for Singapore 2023, the analysis suggests that technological capability is positively associated with productivity, while digital adoption exhibits a more conditional association that becomes more positive only at higher levels of export intensity. The findings therefore support a distinction between firm-internal capability depth and foundational digitally enabled transaction capacity rather than treating both domains as interchangeable aspects of a single digital-capability construct. The study also qualifies the interpretation of the nonlinear I–P literature in this setting. Within the observed export-intensity range, the baseline pattern is better characterized as predominantly positive with mild quadratic curvature than as a formally identified inverted-U, because the implied turning point lies in a sparsely populated upper tail and is imprecisely located. Read in this way, the evidence does not overturn the established I–P literature, nor does it establish a general boundary condition for digitally mature economies; instead, it provides within-context evidence from Singapore on how the conventional nonlinear logic appears in a highly digitalized institutional environment where most firms remain concentrated at low export intensity. More broadly, the study contributes to international business research by sharpening construct interpretation and by showing that technological capability and foundational digital adoption are associated with performance through different empirical patterns. Technological capability shows the more stable positive association with productivity, whereas foundational digital adoption is better interpreted as a conditional scaling resource whose relevance becomes more visible only where cross-border coordination demands are relatively intense. These conclusions are deliberately bounded to the present design and setting, and they point toward the need for comparative and longitudinal research before broader cross-setting claims about how digitalization shapes the I–P pattern can be sustained.

7 Limitations and Future Research

Three limitations warrant caution in interpreting the findings. First, the analysis is based on a single-country cross-section, so the identification scope is associational rather than causal. The observed relationships among technological capability, digital adoption, export intensity, and labour productivity may reflect reverse causation, omitted variables, or productivity-driven selection into exporting and digital adoption. The appropriate interpretation is therefore that the study documents how these variables co-vary in Singapore rather than establishing a causal mechanism. Second, the sample contains a thin right tail of high-intensity exporters. Most firms report zero exports, the 75th percentile of FSTS remains at zero, and only a small fraction of firms occupy the high-export range in which the fitted turning point and the strongest DAI moderation signals appear. The implied turning point in the quadratic specification should therefore be interpreted as a descriptive feature rather than as a structurally identified inflection, because it is estimated from a sparsely populated region and its 95% bootstrap confidence interval is wide ([53%, 253%]) even though an inverted-U shape is recovered in 96.3% of resamples. The same caution applies to the DAI moderation pattern: it remains visible across leave-one-out, trimmedtail, and indicator-sensitivity diagnostics, but precision is necessarily lower in the small exporter subsample (N = 84) and the sparsely populated high-intensity range. The appropriate caution therefore rests on the limited empirical support in the upper tail and the resulting imprecision of tail-based inference. Third, the estimated digital-adoption effect should be interpreted within the measurement boundary of the available indicators. The DAI construct captures Tier 1–2 digital adoption, digital presence and electronic-transaction usage rather than deeper digitally integrated organizational capability or digital dynamic capability; indicator-sensitivity analysis shows that the moderation pattern depends on the foundational digital-infrastructure indicators and weakens when those specific items are dropped. Future research should therefore examine whether the same conditional pattern appears in other digitally advanced economies and under richer measures that capture process integration, organizational digitalization, and higher-order digital capabilities more directly. Panel data, successive enterprise-survey waves, linked administrative records, and stronger quasi-experimental designs, including instrumental-variable or difference-in-differences specifications keyed to digital-infrastructure rollouts or trade-policy shocks, would also help clarify whether the observed associations reflect scaling complementarities, selection effects, or other mechanisms. Concrete instrumental-variable strategies could exploit firm-to-port distance or local customs-clearance times as instruments for export intensity, and regional broadband-coverage shares or within-sector peer-adoption rates as instruments for foundational digital adoption, although Wolfolds and Siegel (2019) caution that such designs require both adequate first-stage strength and credible exclusion restrictions before they can deliver clean causal identification.

Data Availability Statement

The data analyzed in this study are from the World Bank Enterprise Survey (WBES) Singapore 2023 wave, publicly available at www.enterprisesurveys.org. Analysis scripts to replicate the results are available from the corresponding author on reasonable request.

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Conflict of Interest

The authors declare no conflicts of interest.

AI and Data Availability

AI use disclosure: Grammarly was used for grammar and spelling corrections only. No AI tools were used for idea generation, literature synthesis, data analysis, or content creation.

Data availability: The data supporting this study were obtained from the World Bank Enterprise Surveys (WBES) B-READY 2023 Singapore dataset, publicly available at https://www.enterprisesurveys.org. Processed data and replication code are available from the corresponding author upon reasonable request.

References

Aguinis, H., Beaty, J. C., Boik, R. J., & Pierce, C. A. (2005). Effect size and power in assessing moderating effects of categorical variables using multiple regression: A 30-year review. *Journal of Applied Psychology, 90*(1), 94–107. https://doi.org/10.1037/0021-9010.90.1.94 Aiken, L. S., & West, S. G. (1991). *Multiple regression: Testing and interpreting interactions*. Sage. Antonakis, J., Bendahan, S., Jacquart, P., & Lalive, R. (2010). On making causal claims: A review and recommendations. *The Leadership Quarterly, 21*(6), 1086–1120. https://doi.org/10.1016/j.leaqua.2010.10.010 Avenyo, E. K., Tregenna, F., & Kraemer-Mbula, E. (2021). Do productive capabilities affect export performance? Microeconomic evidence from sub-Saharan Africa. *The European Journal of Development Research, 33*(2), 304–329. https://doi.org/10.1057/s41287-020-00328-2 Banalieva, E. R., & Dhanaraj, C. (2019). Internalization theory for the digital economy. *Journal of International Business Studies, 50*(8), 1372–1387. https://doi.org/10.1057/s41267-019-00243-7 Barney, J. (1991). Firm resources and sustained competitive advantage. *Journal of Management, 17*(1), 99–120. https://doi.org/10.1177/014920639101700108 Bharadwaj, A., El Sawy, O. A., Pavlou, P. A., & Venkatraman, N. (2013). Digital business strategy: Toward a next generation of insights. *MIS Quarterly, 37*(2), 471–482. https://doi.org/10.25300/MISQ/2013/37:2.3 Brynjolfsson, E., Rock, D., & Syverson, C. (2021). The productivity J-curve: How intangibles complement general purpose technologies. *American Economic Journal: Macroeconomics, 13*(1), 333–372. https://doi.org/10.1257/mac.20180386 Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Lawrence Erlbaum. Cohen, W. M., & Levinthal, D. A. (1990). Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly, 35*(1), 128–152. https://doi.org/10.2307/2393553 Coltman, T., Devinney, T. M., Midgley, D. F., & Venaik, S. (2008). Formative versus reflective measurement models: Two applications of formative measurement. *Journal of Business Research, 61*(12), 1250–1262. https://doi.org/10.1016/j.jbusres.2008.01.013 Contractor, F. J. (2007). Is international business good for companies? The evolutionary or multi-stage theory of internationalization vs. the transaction cost perspective. *Management International Review, 47*(3), 453–475. https://doi.org/10.1007/s11575-007-0024-2 Contractor, F. J., Kundu, S. K., & Hsu, C.-C. (2003). A three-stage theory of international expansion: The link between multinationality and performance in the service sector. *Journal of International Business Studies, 34*(1), 5–18. https://doi.org/10.1057/palgrave.jibs.8400003 Haans, R. F. J., Pieters, C., & He, Z.-L. (2016). Thinking about U: Theorizing and testing U- and inverted U-shaped relationships in strategy research. *Strategic Management Journal, 37*(7), 1177–1195. https://doi.org/10.1002/smj.2399 Hambrick, D. C., & Mason, P. A. (1984). Upper echelons: The organization as a reflection of its top managers. *Academy of Management Review, 9*(2), 193–206. https://doi.org/10.5465/amr.1984.4277628 Hitt, M. A., Hoskisson, R. E., & Kim, H. (1997). International diversification: Effects on innovation and firm performance in product-diversified firms. *Academy of Management Journal, 40*(4), 767–798. https://doi.org/10.5465/256948 Krammer, S. M. S., Strange, R., & Lashitew, A. (2018). The export performance of emerging economy firms. *International Business Review, 27*(1), 218–230. https://doi.org/10.1016/j.ibusrev.2017.07.003 Lakens, D., Scheel, A. M., & Isager, P. M. (2018). Equivalence testing for psychological research: A tutorial. *Advances in Methods and Practices in Psychological Science, 1*(2), 259–269. https://doi.org/10.1177/2515245918770963 Lall, S. (1992). Technological capabilities and industrialization. *World Development, 20*(2), 165–186. https://doi.org/10.1016/0305-750X(92)90097-F Lind, J. T., & Mehlum, H. (2010). With or without U? The appropriate test for a U-shaped relationship. *Oxford Bulletin of Economics and Statistics, 72*(1), 109–118. https://doi.org/10.1111/j.1468-0084.2009.00569.x Lu, J. W., & Beamish, P. W. (2004). International diversification and firm performance: The S-curve hypothesis. *Academy of Management Journal, 47*(4), 598–609. https://doi.org/10.5465/20159604 MacKinnon, J. G., & White, H. (1985). Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties. *Journal of Econometrics, 29*(3), 305–325. https://doi.org/10.1016/0304-4076(85)90158-7 Marano, V., Arregle, J.-L., Hitt, M. A., Spadafora, E., & van Essen, M. (2016). Home country institutions and the internationalization-performance relationship: A meta-analytic review. *Journal of Management, 42*(5), 1075–1110. https://doi.org/10.1177/0149206315624963 Meyer, K. E., van Witteloostuijn, A., & Beugelsdijk, S. (2017). What's in a p? Reassessing best practices for conducting and reporting hypothesis-testing research. *Journal of International Business Studies, 48*(5), 535–551. https://doi.org/10.1057/s41267-017-0078-8 O'Brien, R. M. (2007). A caution regarding rules of thumb for variance inflation factors. *Quality & Quantity, 41*(5), 673–690. https://doi.org/10.1007/s11135-006-9018-6 Peng, M. W. (2003). Institutional transitions and strategic choices. *Academy of Management Review, 28*(2), 275–296. https://doi.org/10.5465/amr.2003.9416341 Peng, M. W., Wang, D. Y. L., & Jiang, Y. (2008). An institution-based view of international business strategy: A focus on emerging economies. *Journal of International Business Studies, 39*(5), 920–936. https://doi.org/10.1057/palgrave.jibs.8400377 Rugman, A. M., & Verbeke, A. (2004). A perspective on regional and global strategies of multinational enterprises. *Journal of International Business Studies, 35*(1), 3–18. https://doi.org/10.1057/palgrave.jibs.8400073 Shaver, J. M. (2020). Causal identification through a cumulative body of research in the study of strategy and organizations. *Journal of Management, 46*(6), 1244–1256. https://doi.org/10.1177/0149206319846272 Stallkamp, M., & Schotter, A. P. J. (2021). Platforms without borders? The international strategies of digital platform firms. *Global Strategy Journal, 11*(1), 58–80. https://doi.org/10.1002/gsj.1336 Verhoef, P. C., Broekhuizen, T., Bart, Y., Bhattacharya, A., Dong, J. Q., Fabian, N., & Haenlein, M. (2021). Digital transformation: A multidisciplinary reflection and research agenda. *Journal of Business Research, 122*, 889–901. https://doi.org/10.1016/j.jbusres.2019.09.022 Wolfolds, S. E., & Siegel, J. (2019). Misaccounting for endogeneity: The peril of relying on the Heckman two-step method without a valid instrument. *Strategic Management Journal, 40*(3), 432–462. https://doi.org/10.1002/smj.2995

Additional contextual sources (cited in Section 1.1 and Section 5.1)

Enterprise Singapore. (2025, November). *SWITCH 2025 concludes 10th anniversary milestone with record 25,000 attendees* [Press release]. Enterprise Singapore. https://www.enterprisesg.gov.sg/resources/media-centre/media-releases/2025/november/mr05225_switch-2025-concludes-10th-anniversary-milestone-with-record-25000-attendees-strengthens-position-as-premiere-platform-for-deep-tech-innovation

GovTech Singapore. (2024). *Our digital government rankings*. Government Technology Agency of Singapore. https://www.tech.gov.sg/about-us/our-achievements/our-digital-government-rankings/

IMDA. (2025). *Asia Tech x Singapore turns five* [Press release]. Infocomm Media Development Authority. https://www.imda.gov.sg/resources/press-releases-factsheets-and-speeches/press-releases/2025/asia-tech-x-singapore-turns-five

StartupBlink. (2025). *Global Startup Ecosystem Index 2025*. StartupBlink Research. https://www.startupblink.com/startup-ecosystem/singapore

StartupBlink. (2026). *Innovators Business Environment Index 2026*. StartupBlink Research. https://www.startupblink.com/blog/innovators-business-environment-index/ World Bank. (2024). *World Bank Enterprise Survey: Singapore 2023*. https://www.enterprisesurveys.org